

Documentation Templates

Purpose

Several stages of the Fairness Intervention Playbook require formal documentation artifacts.

To ensure consistency, usability, and audit readiness, standardized checklists are provided below.

All required boxes must be completed before deployment.

The example scenario included at the end demonstrates how these templates are applied in practice.

All metric values shown in the example are illustrative and used for demonstration purposes only.

Workflow Integration

These templates are used across multiple stages of the Fairness Implementation Playbook, including intervention selection, implementation, validation, and governance.

Completion of relevant templates is mandatory before progressing to the next stage in the workflow.

Decision Traceability Requirement

Each checklist must document:

- Identified bias stage
- Selected intervention type
- Justification for selection
- Explanation for skipped stages
- Risk classification reference

This ensures that all decisions are traceable to causal analysis and intervention selection.

1. Pre-Processing – Dataset Change Log Checklist

Owner: _____

Date: _____

Model Version: _____

Fairness Issue Identification

- ☐ Disparity identified and documented
- ☐ Affected group(s) specified
- ☐ Supporting metrics included

Type of Data Change

- ☐ Rebalancing performed
- ☐ Reweighting applied
- ☐ Biased labels corrected
- ☐ Low-quality records removed
- ☐ Measurement normalization applied
- ☐ Other (specified): _____

Change Description

- ☐ Records affected quantified
- ☐ Features modified documented
- ☐ Rationale clearly stated

Risk Assessment

- ☐ Impact on accuracy evaluated
- ☐ Impact on other groups evaluated
- ☐ Distribution shift assessed

Approval

- ☐ Data Lead approval
- ☐ Risk & Compliance approval

2. In-Processing – Model Constraint Summary Checklist

Owner: _____

Model Version: _____

Date: _____

Bias Identification

- ☐ Persistent disparity documented
- ☐ Metrics provided

Intervention Type

- ☐ Fairness constraint added
- ☐ Loss function modified
- ☐ Proxy feature limited
- ☐ Calibration adjusted
- ☐ Other (specified): _____

Implementation

- ☐ Training changes documented
- ☐ Hyperparameters recorded
- ☐ Version control updated

Trade-Off Evaluation

- ☐ Fairness improvement quantified
- ☐ Performance impact measured
- ☐ Business impact assessed

Stability

- ☐ Retraining stability tested

Approval

- ☐ ML Lead approval
- ☐ Risk & Compliance approval

3. In-Processing – Trade-Off Explanation Checklist

Owner: _____

Date: _____

Fairness Impact

- ☐ Metric improvement documented
- ☐ Disparity reduction quantified

Performance Impact

- ☐ Accuracy change recorded
- ☐ Precision/Recall change recorded
- ☐ KPI impact assessed

Governance Decision

- ☐ Within acceptable thresholds
- ☐ Escalation required
- ☐ Approved

4. Post-Processing – Threshold Adjustment Checklist

Owner: _____

Date: _____

Justification

- ☐ Bias concentrated at threshold identified
- ☐ Urgency of harm documented

Threshold Change

- ☐ Original threshold recorded
- ☐ New threshold recorded
- ☐ Affected group(s) specified

Before / After Comparison

- ☐ Error rate comparison documented
- ☐ Outcome disparity comparison documented

Operational Impact

- ☐ Manual review volume assessed
- ☐ Processing time impact assessed

Monitoring Plan

- ☐ Ongoing monitoring defined

Approval

- ☐ Product approval
- ☐ Risk & Compliance approval

5. Validation & Monitoring Checklist

Owner: _____

Date: _____

Fairness Metrics

- ☐ Error-based metrics evaluated
- ☐ Outcome-based metrics evaluated
- ☐ Stability metrics evaluated

Performance Metrics

- ☐ Accuracy assessed
- ☐ KPI impact assessed

Decision

- ☐ Pass
- ☐ Fail
- ☐ Escalate

Final Approval

- ☐ Risk & Compliance sign-off

Connection to Validation Framework

All checklist outputs must be validated using the Validation and Monitoring Framework.

No checklist completion is considered sufficient unless validation confirms:

- Fairness improvement
- Acceptable performance impact
- Stability across groups and time

Validation results must be attached to the final report.

6. Governance Rule

- ☐ All required checklists completed
- ☐ All required approvals obtained
- ☐ Validation passed
- ☐ Documentation stored in project record

Deployment is strictly prohibited unless all conditions above are satisfied.

If it is not documented, it is not approved.

Example

Scenario 1 – High-Risk System

System: IVRAS

Risk Level: High-Risk

Causal analysis shows bias at data, model, and decision stages.

Pre-Processing

- ☒ Disparity identified
- ☒ Data imbalance corrected
- ☒ Measurement normalization applied
- ☒ Accuracy impact evaluated
- ☒ Risk & Compliance approval

In-Processing

- ☒ Persistent disparity after data fix
- ☒ Fairness constraint added
- ☒ Trade-offs evaluated
- ☒ Stability tested
- ☒ Risk & Compliance approval

Post-Processing

- ☒ Threshold sensitivity identified
- ☒ Threshold adjusted
- ☒ Monitoring plan created
- ☒ Risk & Compliance approval

Validation

- ☒ Fairness metrics improved
- ☒ Performance within approved limits
- ☒ Executive approval (required for High-Risk)

Result

All stages were evaluated and applied.
A layered intervention is required due to multi-stage bias.
Full governance approval is mandatory.

Scenario 2 – Lower-Risk System

System: News Recommendation Model
Risk Level: Lower-Risk

Causal analysis shows bias only at data stage.

Pre-Processing

- ☒ Disparity identified
- ☒ Dataset rebalanced
- ☒ Performance verified
- ☒ Risk & Compliance approval

In-Processing

- ☐ Model-stage bias detected
- ☐ Constraint required

Justification:
No model amplification observed.
Stage skipped and documented.

Post-Processing

- ☐ Threshold harm detected
- ☐ Adjustment required

Justification:
No decision-stage disparity.
Stage skipped and documented.

Validation

- ☑ Fairness metrics improved
- ☑ Performance unaffected
- ☑ Risk & Compliance sign-off

Result

Only the necessary stage applied.
Skipping permitted due to lower risk and isolated bias.
Proportionate validation sufficient.

These templates ensure that fairness interventions are consistently applied, documented, validated, and governed across the entire AI lifecycle.